

BIOSTAT 724 - Homework 1

Austin Allen

Problem 1

In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and $1 - p$ the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question, given that he or she answered it correctly? What happens when m is very large? What happens when p is close to 0?

Let's solve this using Baye's Theorem. First, it'll be helpful to define some terms:

- $P(\text{Knows}) = p$
- $P(\text{Guess}) = 1 - p$
- $P(\text{Correct} \mid \text{Guess}) = \frac{1}{m}$
- We want to know $P(\text{Knows} \mid \text{Correct})$

$$\begin{aligned} P(\text{Knows} \mid \text{Correct}) &= \frac{P(\text{Correct} \mid \text{Knows})P(\text{Knows})}{P(\text{Correct} \mid \text{Knows})P(\text{Knows}) + P(\text{Correct} \mid \text{Guess})P(\text{Guess})} \\ &= \frac{(1)(p)}{(1)(p) + (\frac{1}{m})(1 - p)} \\ &= \frac{p}{p + \frac{1-p}{m}} \end{aligned}$$

We can see from this result that as m gets arbitrarily large, the denominator approaches p and the probability goes to 1. This aligns with our intuition because a correct guess among a larger number of choices would be less likely. Likewise, if p is close to 0, the probability is also close to zero.

Problem 2

Suppose urn H is filled with 40% green balls and 60% red balls, and urn T is filled with 60% green balls and 40% red balls. Someone will flip a coin and then select a ball from urn H or urn T depending on whether the coin lands heads or tails, respectively. Let X be 1 or 0 if the coin lands heads or tails, and let Y be 1 or 0 if the ball is green or red.

Part a

Write out the joint distribution of X and Y in a table.

```
# | echo: false
import pandas as pd

# Define the joint probability table
df = pd.DataFrame(
    {0: [0.2, 0.3], 1: [0.3, 0.2]},
    index=pd.Index([0, 1], name="Y"),
    columns=pd.Index([0, 1], name="X"),
)

# Add marginal probabilities
df["P(Y)"] = df.sum(axis=1) # Row sums (marginal P(Y))
df.loc["P(X)"] = df.sum(axis=0) # Column sums (marginal P(X))

# Display the table
print(df)
```

X	0	1	P(Y)
Y			
0	0.2	0.3	0.5
1	0.3	0.2	0.5
P(X)	0.5	0.5	1.0

Part b

Find $E(Y)$. What is the probability that the ball is green?

$E(Y) = \sum y \cdot f_Y(y)$. In our table, we see that Y can take on two values, 0 and 1. We can use the marginal probability to find that $\sum y \cdot f_Y(y) = (0)(0.5) + (1)(0.5) = 0.5$. Thus, $P(\text{Ball is green}) = 0.5$.